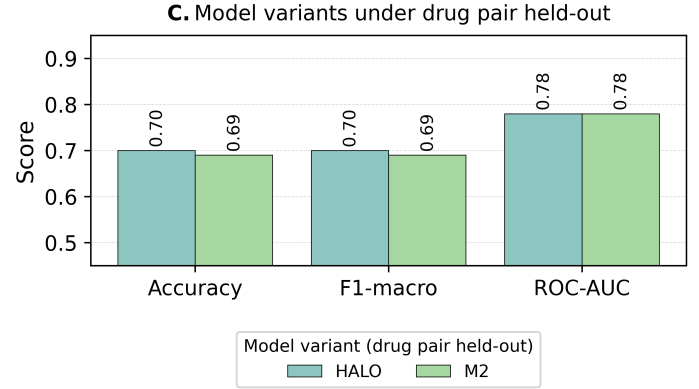
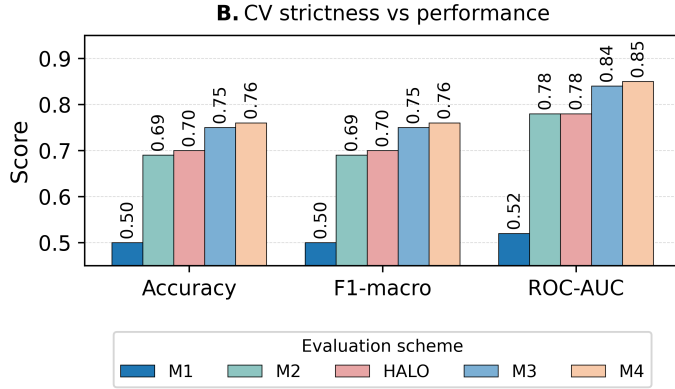


A. Model configuration, evaluation scheme, and performance.

Model	Configuration		Evaluation	Performance		
	Strain features	Feature selection		Acc.	AUC	F1
M1	Yes	Nested (within CV folds)	Drug pair + strain held-out	0.50	0.52	0.50
M2	Yes	Nested (within CV folds)	Drug pair held-out	0.69	0.78	0.69
HALO	No	Nested (within CV folds)	Drug pair held-out	0.70	0.78	0.70
M3	No	Global (pre-CV)	Drug pair held-out	0.75	0.84	0.75
M4	No	Training-set only	Standard stratified split	0.76	0.85	0.76



D. Nested cross validation scheme

